

# Natural Language Processing (CS24351) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

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Module I: Linguistic Foundations & Preprocessing — Topics Covered

1. Five Levels of Linguistic Analysis
2. Ambiguity in Natural Language (4 Forms)
3. Tokenization & Byte-Pair Encoding (BPE)
4. Porter Stemmer vs. WordNet Lemmatization
5. Regular Expressions for NLP Information Extraction

1. Five Levels of Linguistic Analysis

| Linguistic Level          | Core Domain of Study                                                | Example Task / Challenge                                  |
|---------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|
| 1. Phonetics / Phonology  | Acoustic sounds, phonemes, and pronunciation rules.                 | Speech-to-Text acoustic modeling.                         |
| 2. Morphology             | Structure of words, morphemes, prefixes, suffixes, and inflections. | Stemming ('running' → 'run').                             |
| 3. Syntax                 | Grammatical structure, phrase groupings, and word order.            | POS Tagging and Parse Trees.                              |
| 4. Semantics              | Literal meaning of words, phrases, and sentence composition.        | Word Sense Disambiguation (WSD).                          |
| 5. Pragmatics / Discourse | Contextual meaning, speaker intent, coreference, and sarcasm.       | Anaphora resolution ('Mary saw the car. She bought it.'). |

## Module II: Language Models, Smoothing & Parsing — Topics Covered

1.  $N$ -gram Language Models & Markov Assumption
2. Perplexity Evaluation Metric
3. Smoothing: Laplace, Good-Turing, Kneser-Ney
4. HMM POS Tagging & The Viterbi Algorithm
5. Probabilistic Context-Free Grammars (PCFGs)

### 1. $N$ -gram Language Models & Perplexity

$$P(w_1, w_2, \dots, w_N) = \prod_{k=1}^N P(w_k \mid w_{k-n+1}^{k-1})$$

$$\text{Perplexity}(W) = P(w_1, \dots, w_N)^{-1/N} = 2^{-\frac{1}{N} \sum \log_2 P(w_i \mid w_{i-n+1}^{i-1})}$$

### 2. Smoothing Techniques Comparison

- **Laplace (Add-1) Smoothing:**  $P(w_n \mid w_{n-1}) = \frac{C(w_{n-1}, w_n) + 1}{C(w_{n-1}) + V}$ . (Tends to allocate too much probability mass to unseen items).
- **Kneser-Ney Smoothing:** State-of-the-art  $N$ -gram smoothing combining absolute discounting with continuation probabilities based on how versatile a word is as a novel continuation.

Module III: Lexical Semantics & Word Embeddings — Topics Covered

1. WordNet Synsets & Lesk Disambiguation
2. TF-IDF Vector Space & Cosine Similarity
3. Word2Vec: Continuous Bag of Words (CBOW)
4. Word2Vec: Skip-Gram with Negative Sampling
5. GloVe Global Vectors & FastText Subwords

1. Word2Vec Architectures

| Architecture                            | Objective Formulation                                                                                                                                         | Key Strength                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Continuous Bag-of-Words (CBOW)          | Predicts target center word $w_t$ given sum of surrounding context words $w_{t-c}, \dots, w_{t+c}$ .                                                          | Faster to train; better representation for frequent words.     |
| Skip-Gram with Negative Sampling (SGNS) | Predicts surrounding context words given target center word $w_t$ : $\max \sum \log \sigma(v'_c \cdot v_w) + \sum \mathbb{E}[\log \sigma(-v'_k \cdot v_w)]$ . | Superior performance on rare words and fine-grained analogies. |

## Module IV: Neural Sequence Models & Transformers — Topics Covered

1. Recurrent Neural Networks (RNN) & BPTT
2. Long Short-Term Memory (LSTM) & GRU Gating
3. Seq2Seq Architecture & Attention Mechanisms
4. The Transformer: Scaled Dot-Product & Multi-Head
5. Pretrained Models: BERT (MLM) vs. GPT (Autoregressive)

### 1. The Transformer Attention Formulation (Vaswani et al., 2017)

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad \text{where } \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

#### BIT Mesra Exam Question (10 Marks)

**Problem:** Compare BERT and GPT architectures in terms of attention directionality, pre-training objectives, and downstream fine-tuning.

**Solution:**

- **BERT:** Bidirectional Transformer Encoder. Pretrained on Masked Language Modeling (MLM 15% tokens) + Next Sentence Prediction (NSP). Ideal for classification, NER, QA extraction.
- **GPT:** Unidirectional Autoregressive Decoder (Causal attention mask). Pretrained on Next-Token Prediction. Ideal for text generation, creative writing, and in-context few-shot prompting.

## Module V: NLP Applications & Responsible AI — Topics Covered

1. Text Classification & Sentiment Analysis
2. Named Entity Recognition (BiLSTM-CRF)
3. Machine Translation Paradigms & BLEU Score
4. Task-Oriented Dialogue Systems
5. Fairness, Societal Bias & Ethics in NLP

### 1. Machine Translation Evaluation: BLEU Metric

**BLEU (Bilingual Evaluation Understudy)** computes modified  $n$ -gram precision penalized by brevity:

$$\text{BLEU} = \text{BP} \times \exp \left( \sum_{n=1}^N w_n \log p_n \right)$$
$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases}$$

where  $c$  is candidate translation length and  $r$  is reference corpus length.